



Topic 16

Electrical Energy and Capacitance

Reading Question 16-1 (1 of 2)

A positive charge in a uniform electric field accelerates, gaining 5 J of kinetic energy. How does the charge's potential energy change?

1. It loses an unknown amount.
2. It doesn't change.
3. It gains 5 J.
4. It loses 5 J.

Reading Question 16-1 (2 of 2)

A positive charge in a uniform electric field accelerates, gaining 5 J of kinetic energy. How does the charge's potential energy change?

1. It loses an unknown amount.
2. It doesn't change.
3. It gains 5 J.
4. **Answer: It loses 5 J.**

Reading Question 16-2 (1 of 2)

A and B are both points on an equipotential surface. What is the potential difference between point A and B?

1. To solve this, the electric potential of the equipotential surface must be known.
2. It is zero.
3. To solve this, the electric potential at both A and B must be known.
4. To solve this, the path taken from A to B must be known.

Reading Question 16-2 (2 of 2)

A and B are both points on an equipotential surface. What is the potential difference between point A and B?

1. To solve this, the electric potential of the equipotential surface must be known.
- 2. Answer: It is zero.**
3. To solve this, the electric potential at both A and B must be known.
4. To solve this, the path taken from A to B must be known.

Reading Question 16-3 (1 of 2)

As one moves away from a positive point charge, the electric potential changes in a way that cannot be determined from the information given.

1. increases.
2. decreases.
3. remains the same.
4. changes in ways we cannot determine with the given information.

Reading Question 16-3 (2 of 2)

As one moves away from a positive point charge, the electric potential changes in a way that cannot be determined from the information given.

1. increases.
2. **Answer: decreases.**
3. remains the same.
4. changes in ways we cannot determine with the given information.

Reading Question 16-4 (1 of 2)

Capacitance is

1. measured in farads.
2. the ratio of the magnitude of the charge on either conductor of a capacitor to the magnitude of the potential difference between the conductors.
3. constant for a parallel-plate capacitor.
4. all of these choices.

Reading Question 16-4 (2 of 2)

Capacitance is

1. measured in farads.
2. the ratio of the magnitude of the charge on either conductor of a capacitor to the magnitude of the potential difference between the conductors.
3. constant for a parallel-plate capacitor.
4. **Answer: all of these choices.**

Reading Question 16-5 (1 of 2)

To increase the capacitance of a parallel-plate capacitor, you can

1. increase the area of the plates.
2. increase the distance between the plates.
3. all of these choices.
4. none of these choices.

Reading Question 16-5 (2 of 2)

To increase the capacitance of a parallel-plate capacitor, you can

1. **Answer: increase the area of the plates.**
2. increase the distance between the plates.
3. all of these choices.
4. none of these choices.

Reading Question 16-6 (1 of 2)

Select the option that best describes a dielectric.

1. A dielectric is a nonconducting material.
2. A dielectric is the material when placed between the plates of a capacitor will increase the electric field.
3. A dielectric is something that when placed between the plates decreases the capacitance of a capacitor.
4. All of these choices are true.

Reading Question 16-6 (2 of 2)

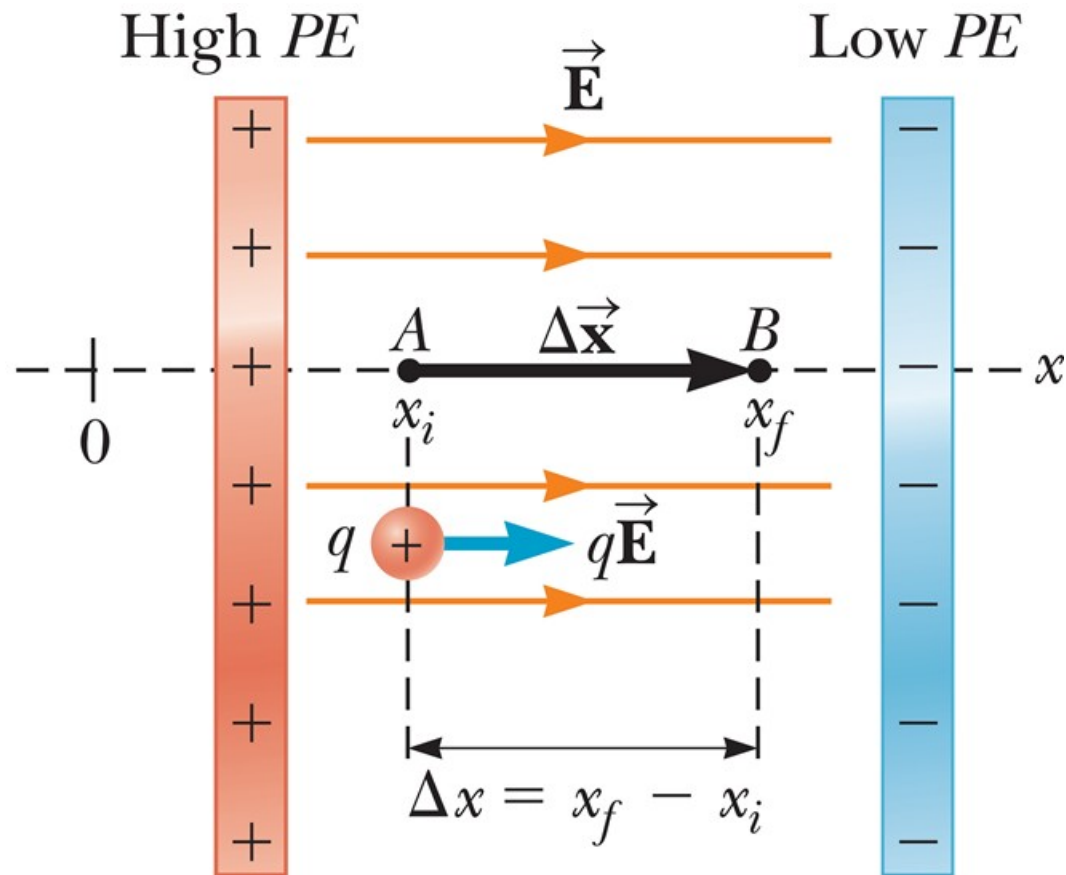
Select the option that best describes a dielectric.

- 1. Answer: A dielectric is a nonconducting material.**
2. A dielectric is the material when placed between the plates of a capacitor will increase the electric field.
3. A dielectric is something that when placed between the plates decreases the capacitance of a capacitor.
4. All of these choices are true.

Topic 16: Electrical Energy and Capacitance

TOPIC DISCUSSION

Work and Electric Potential Energy (1 of 4)

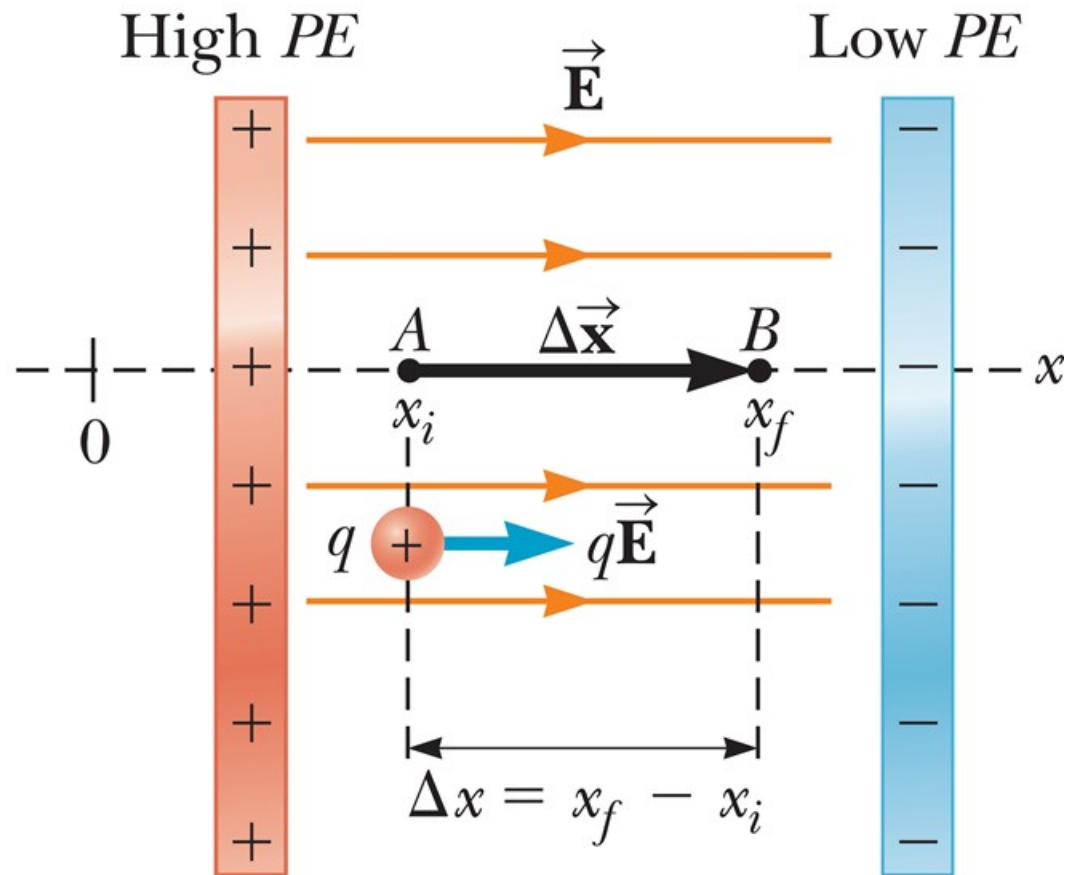


$$\Delta PE = PE_f - PE_i = -W_F$$

$$W_{AB} = F_x \Delta x = qE_x(x_f - x_i)$$

$$|W| = |q|Ed \quad (\text{magnitude})$$

Work and Electric Potential Energy (2 of 4)

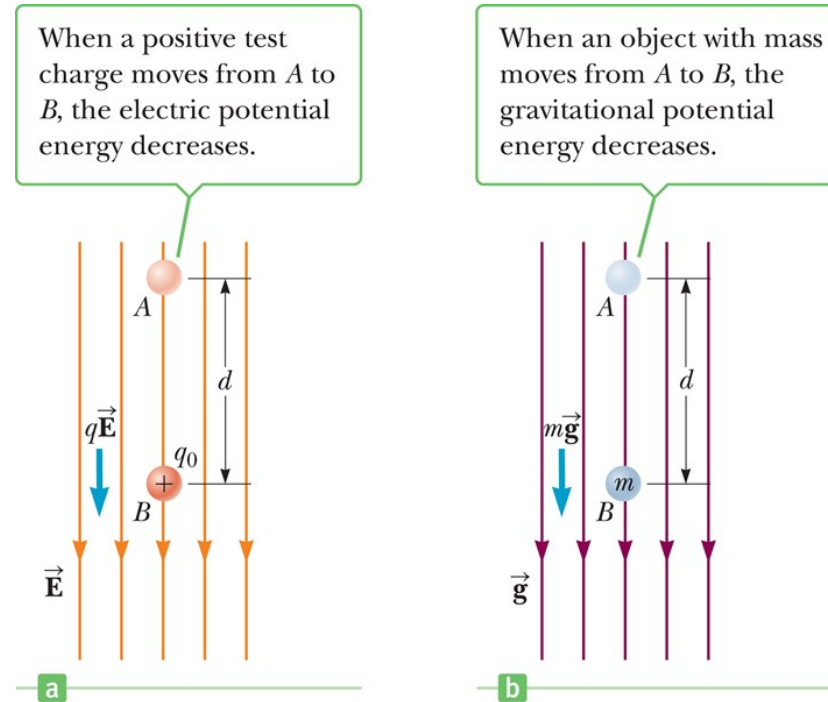


$$W = qE_x \Delta x = \Delta KE$$

$$\Delta PE = -W_{AB}$$

$$= -qE_x \Delta x \quad \text{SI unit: J}$$

Work and Electric Potential Energy (3 of 4)

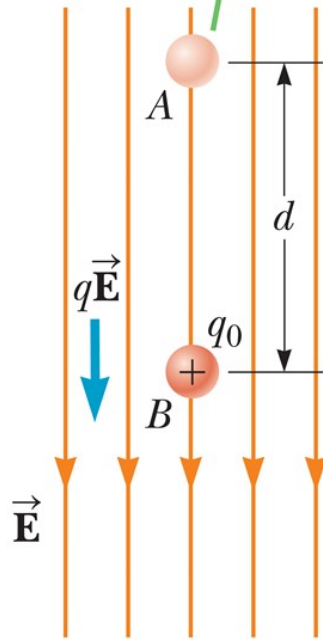


$$\Delta KE + \Delta PE_{el} = \Delta KE + (0 - |q|Ed) = 0 \quad \rightarrow \quad \Delta KE = |q|Ed$$

$$\Delta KE + \Delta PE_g = \Delta KE + (0 - mgd) = 0 \quad \rightarrow \quad \Delta KE = mgd$$

Work and Electric Potential Energy (4 of 4)

When a positive test charge moves from A to B , the electric potential energy decreases.



$$W = \frac{1}{2}mv^2 + |q|Ed$$

Think – Pair – Share 1

If an electron is released from rest in a uniform electric field, the electric potential energy of the charge-field system

1. increases.
2. decreases.
3. remains the same.

Electric Potential (1 of 2)

$$\Delta V = V_B - V_A = \frac{\Delta PE}{q} \quad \text{SI unit: J/C or V}$$

$$\frac{\Delta PE}{q} = \frac{-qE_x \Delta x}{q} = -E_x \Delta x$$

$$\Delta V_x = -E_x \Delta x$$

$$1 \text{ N/C} = 1 \text{ V/m}$$

Think – Pair – Share 2

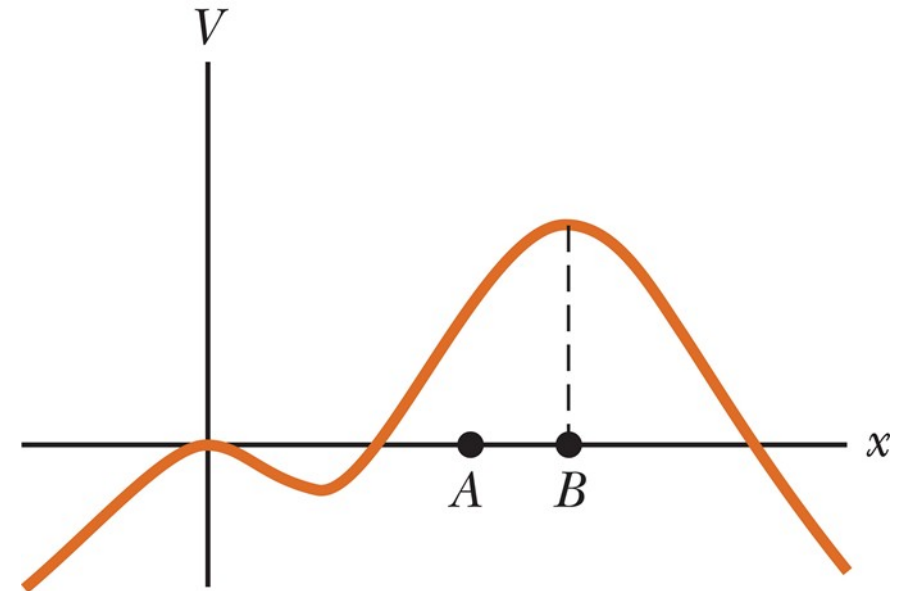
If a negatively charged particle is placed at rest in an electric potential field that increases in the positive x -direction, the particle will

1. accelerate in the positive x -direction.
2. accelerate in the negative x -direction.
3. remain at rest

Think – Pair – Share 3

The figure below is a graph of an electric potential as a function of position. If a positively charged particle is placed at point A , what will its subsequent motion be? It will

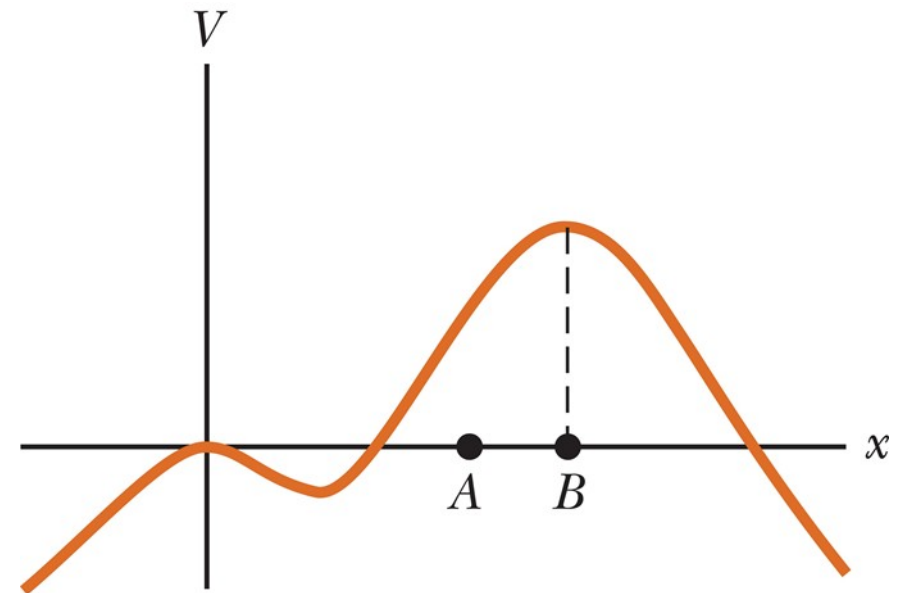
1. go to the right.
2. go to the left.
3. remain at point A .
4. oscillate around point B .



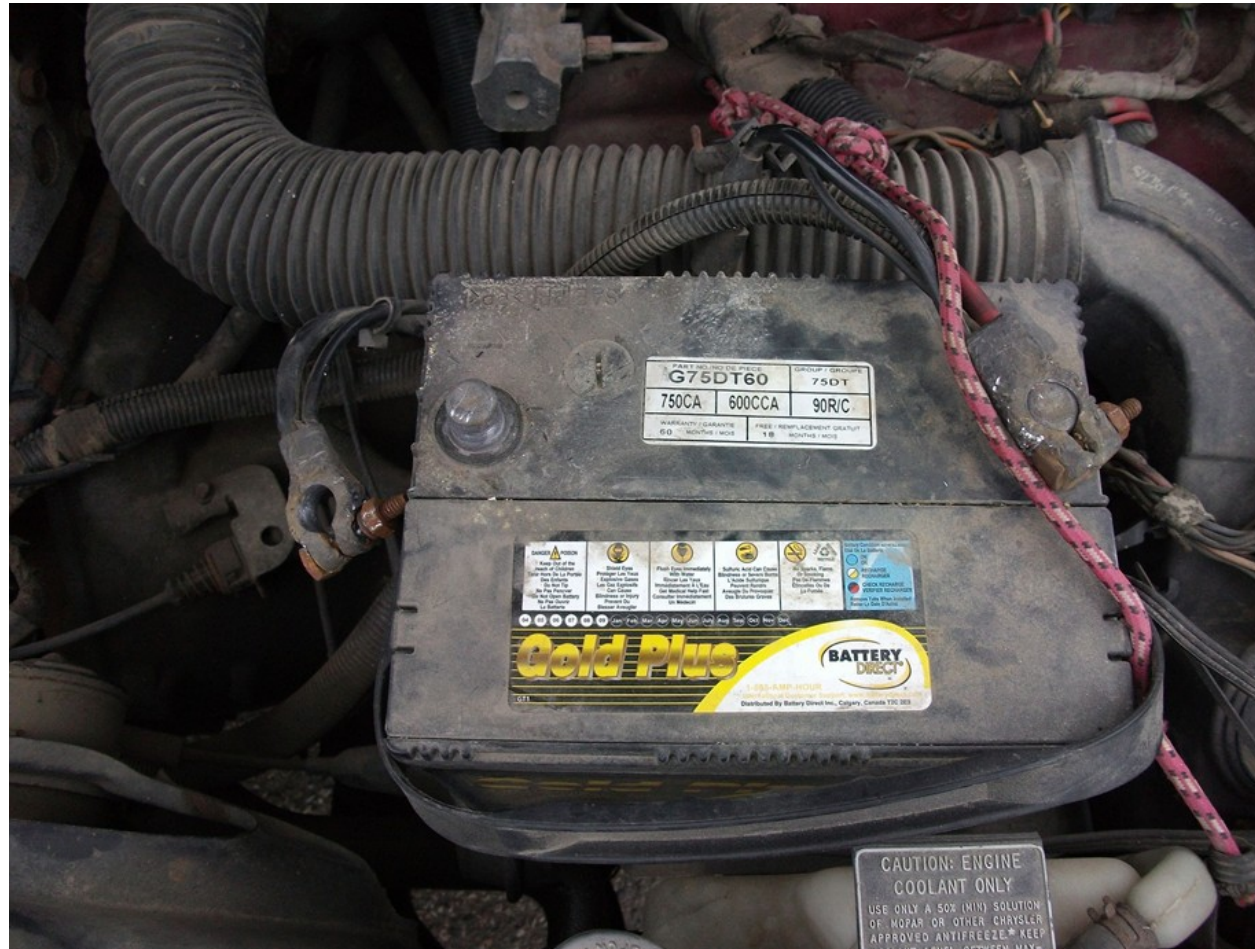
Think – Pair – Share 4

If a negatively charged particle is placed at point B in the figure below and given a very small kick to the right, what will its subsequent motion be? It will

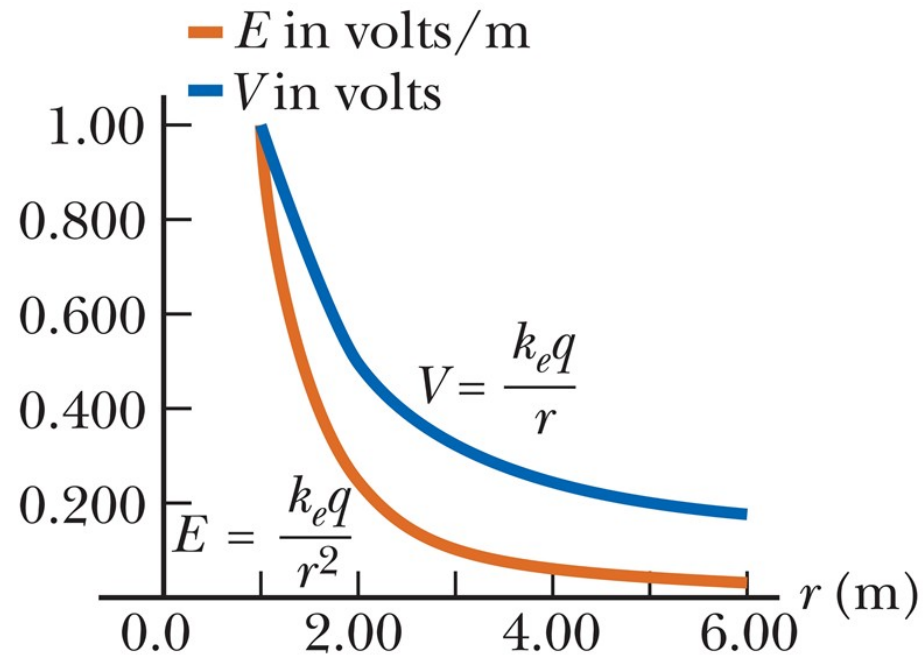
1. go to the right and not return.
2. go to the left.
3. remain at point B .
4. oscillate around point B .



Electric Potential (2 of 2)



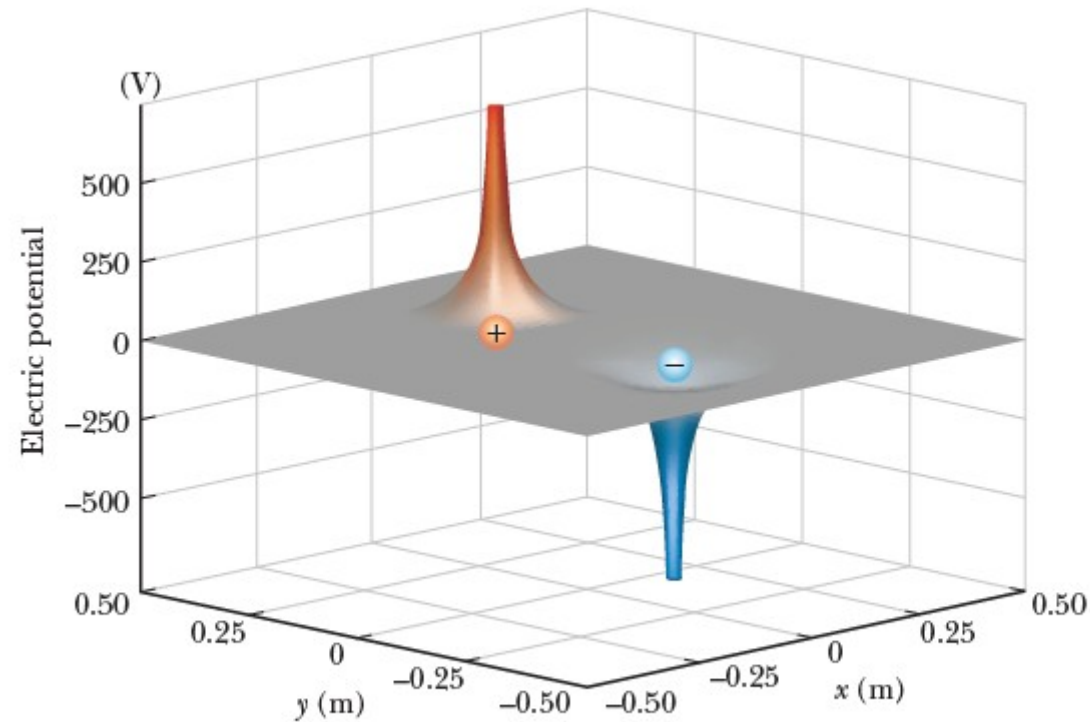
Electric Potential and Potential Energy Due to Point Charges (1 of 3)



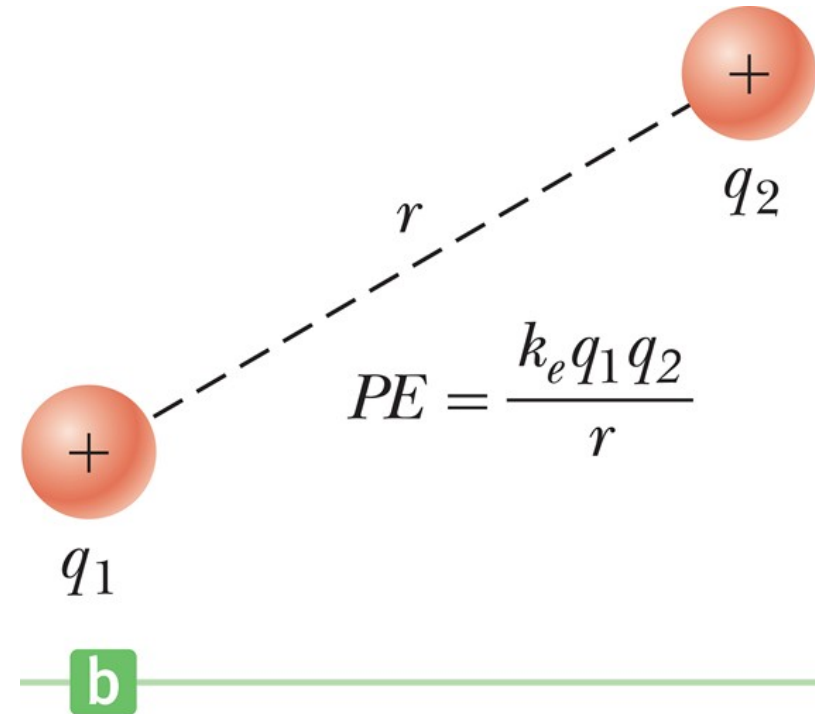
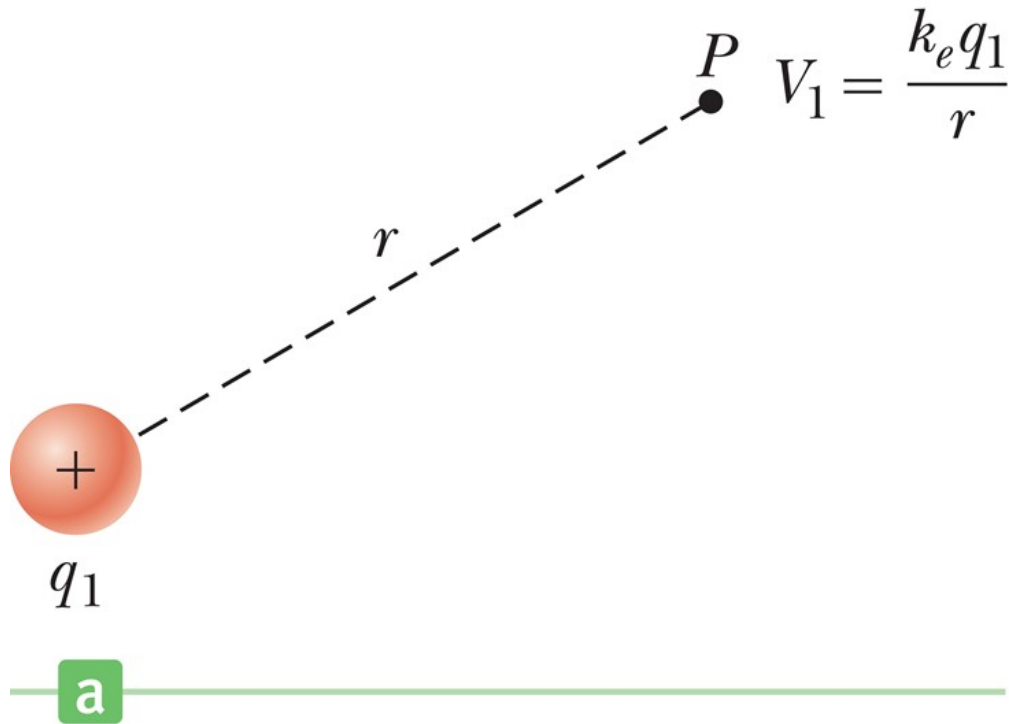
$$V = k_e \frac{q}{r}$$

Superposition principle: the total electric potential P due to several point charges is the algebraic sum of the electric potentials due to the individual charges.

Electric Potential and Potential Energy Due to Point Charges (2 of 3)



Electric Potential and Potential Energy Due to Point Charges (3 of 3)



Think – Pair – Share 5

Consider a collection of charges in a given region and suppose all other charges are distant and have a negligible effect. Further, the electric potential is taken to be zero at infinity. If the electric potential at a given point in the region is zero, which of the following statements must be true?

- (a) The electric field is zero at that point.
- (b) The electric potential energy is a minimum at that point.
- (c) There is no net charge in the region.
- (d) Some charges in the region are positive, and some are negative.
- (e) The charges have the same sign and are symmetrically arranged around the given point.

- 1. (a) only
- 2. (d) only
- 3. (b), (c), and (d)
- 4. (a) and (e)

Think – Pair – Share 6

A spherical balloon contains a positively charged particle at its center. True or false, as the balloon is inflated to a larger volume while the charged particle remains at the center, which of the following is true?

1. The electric potential at the surface of the balloon increases.
2. The magnitude of the electric field at the surface of the balloon increases.
3. The electric flux through the balloon remains the same.
4. none of these

Problem-Solving Strategy: Electric Potential

1. Draw a diagram
2. Calculate and label distances
3. For each charge, calculate:

$$V = k_e \frac{q}{r}$$

4. Sum to get the electric potential

Potentials and Charged Conductors (1 of 2)

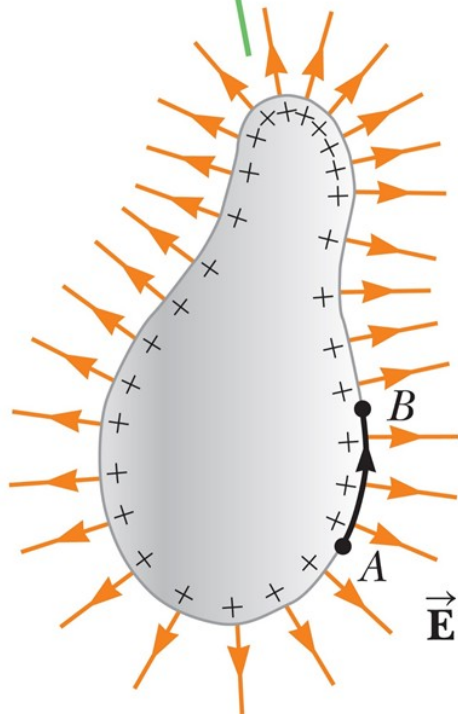
$$W = -\Delta PE$$

$$\Delta PE = q(V_B - V_A)$$

$$W = -q(V_B - V_A)$$

Potentials and Charged Conductors (2 of 2)

Notice from the spacing of the positive signs that the surface charge density is nonuniform.



$$W = -q(V_B - V_A)$$

- The electric potential is a constant everywhere on the surface of a charged conductor in equilibrium.
- The electric potential is constant everywhere inside a conductor and equal to that same value at the surface.

The Electron Volt

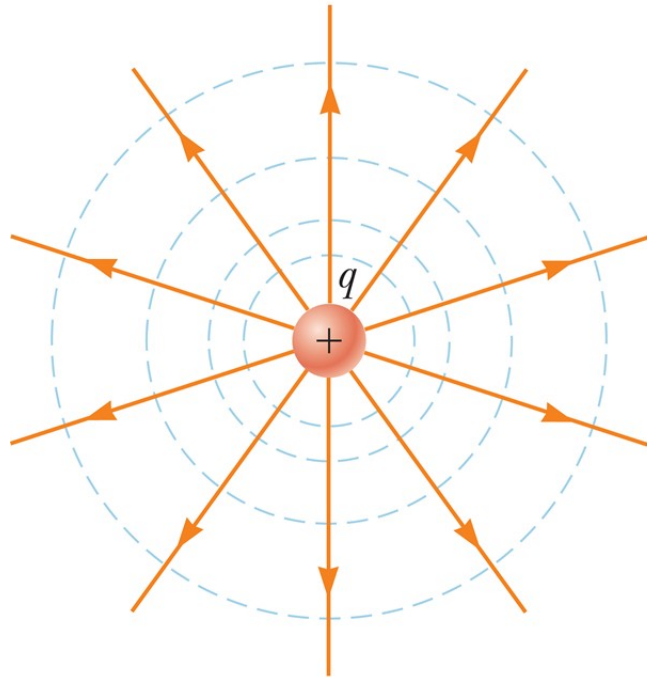
$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ C} \cdot \text{V} = 1.60 \times 10^{-19} \text{ J}$$

Think – Pair – Share 7

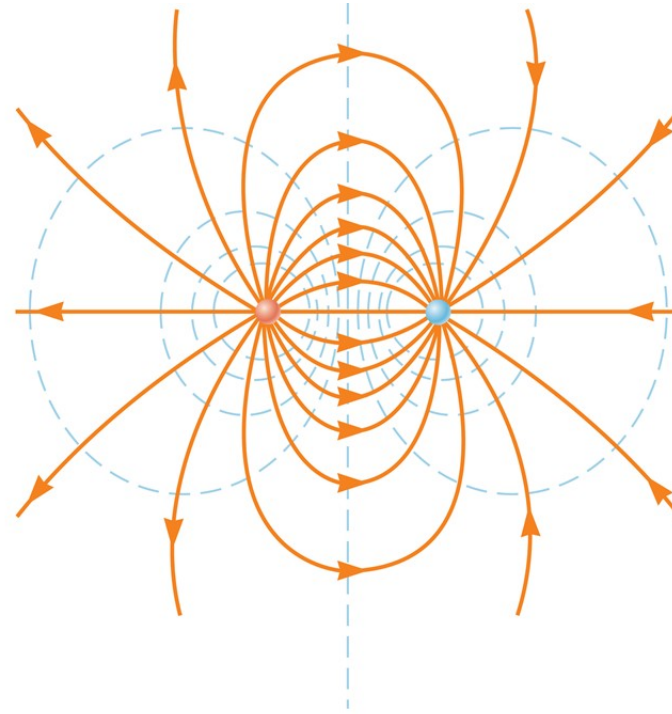
An electron, initially at rest, accelerates through a potential difference of 1 V, gaining kinetic energy KE_e , while a proton, also initially at rest, accelerates through a potential difference of -1 V, gaining kinetic energy KE_p , which of the following relationships holds?

1. $KE_e = KE_p$
2. $KE_e < KE_p$
3. $KE_e > KE_p$
4. The answer can't be determined from the given information.

Equipotential Surfaces



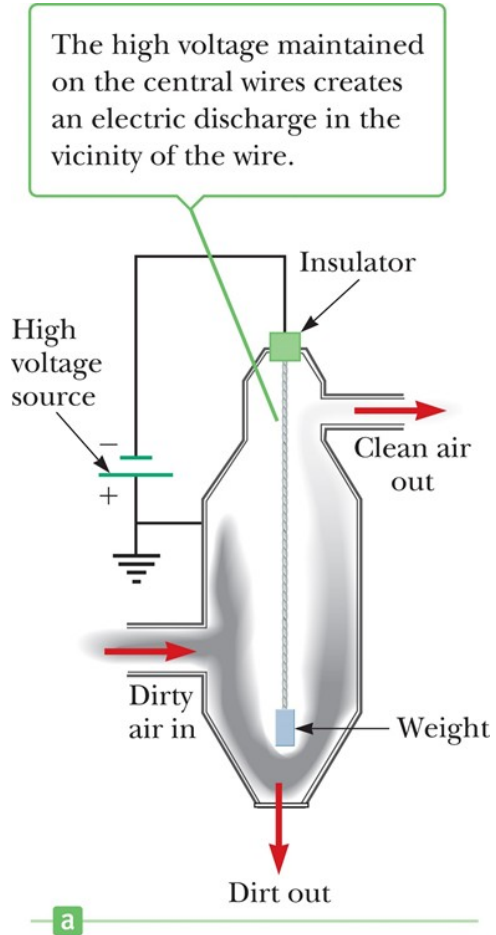
a



b

$$V = k_e \frac{q}{r}$$

The Electrostatic Precipitator



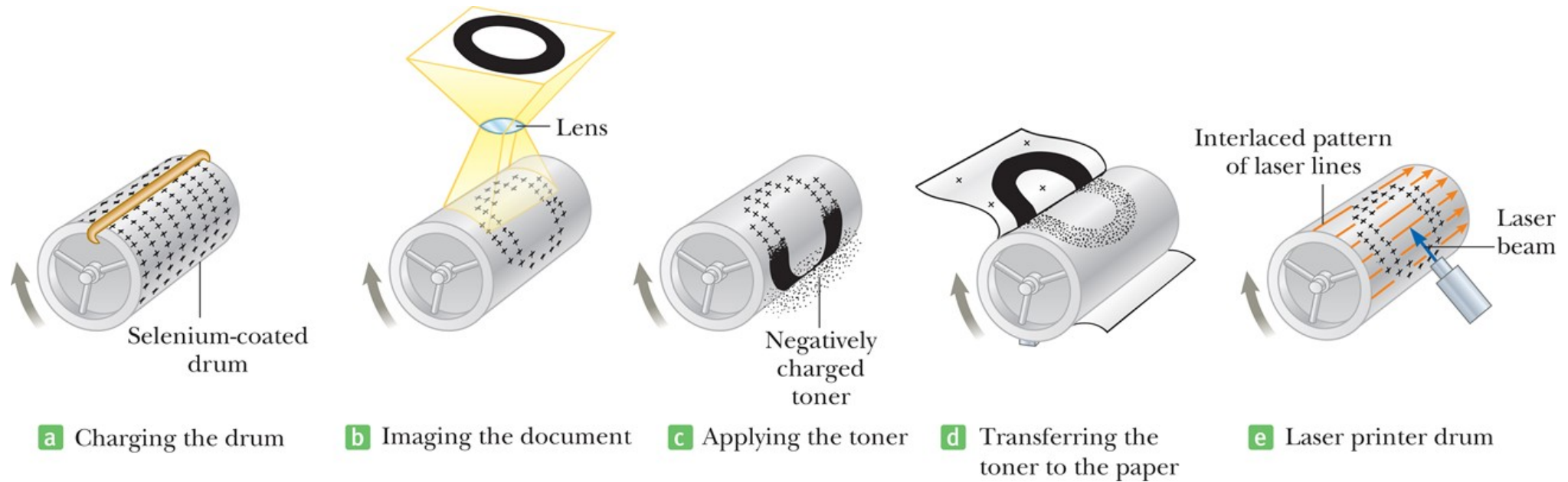
Precipitator operating



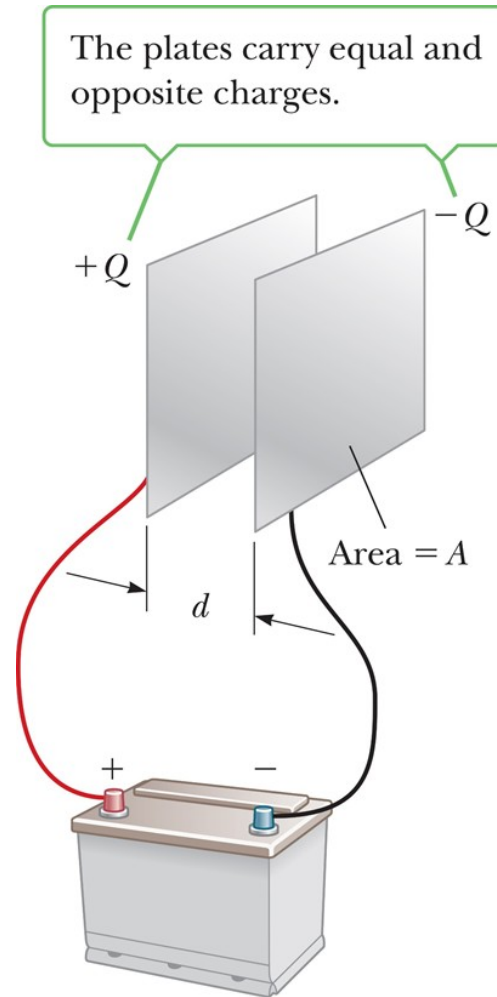
Precipitator not operating



Xerography and Laser Printers



Capacitors (1 of 2)

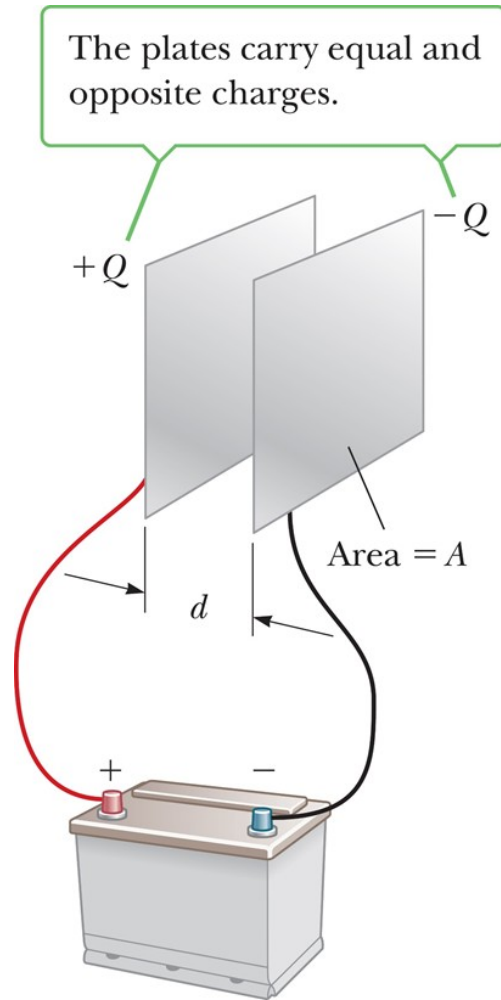


Capacitors (2 of 2)

$$C \equiv \frac{Q}{\Delta V} \quad \text{SI unit: C/V}$$

$$Q = C \Delta V = (3.0 \times 10^{-6} \text{ F})(12 \text{ V}) = 36 \mu\text{C}$$

The Parallel-Plate Capacitor (1 of 3)

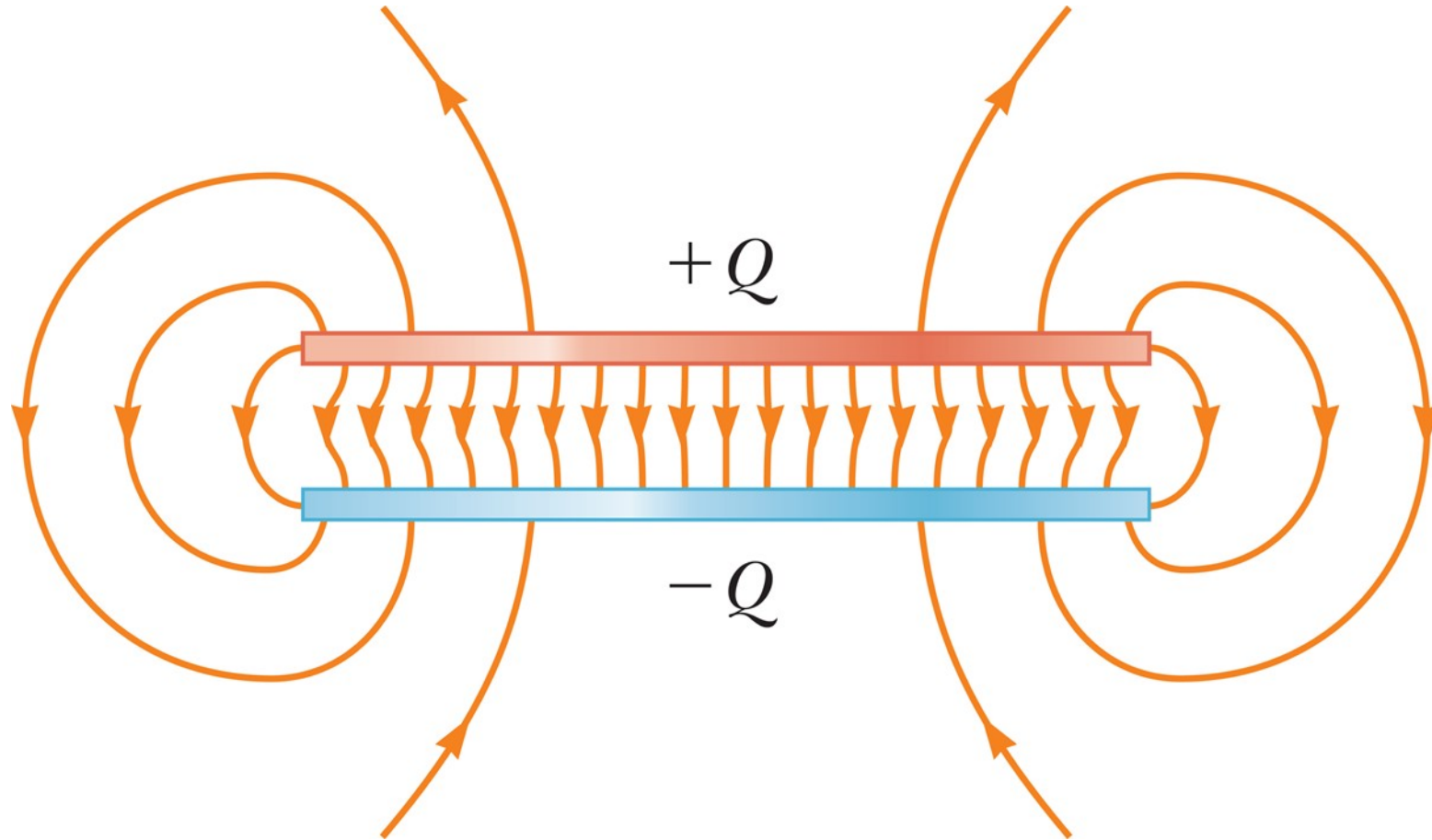


$$E = \frac{\sigma}{\epsilon_0} \quad \Delta V = Ed \quad Q = \sigma A$$

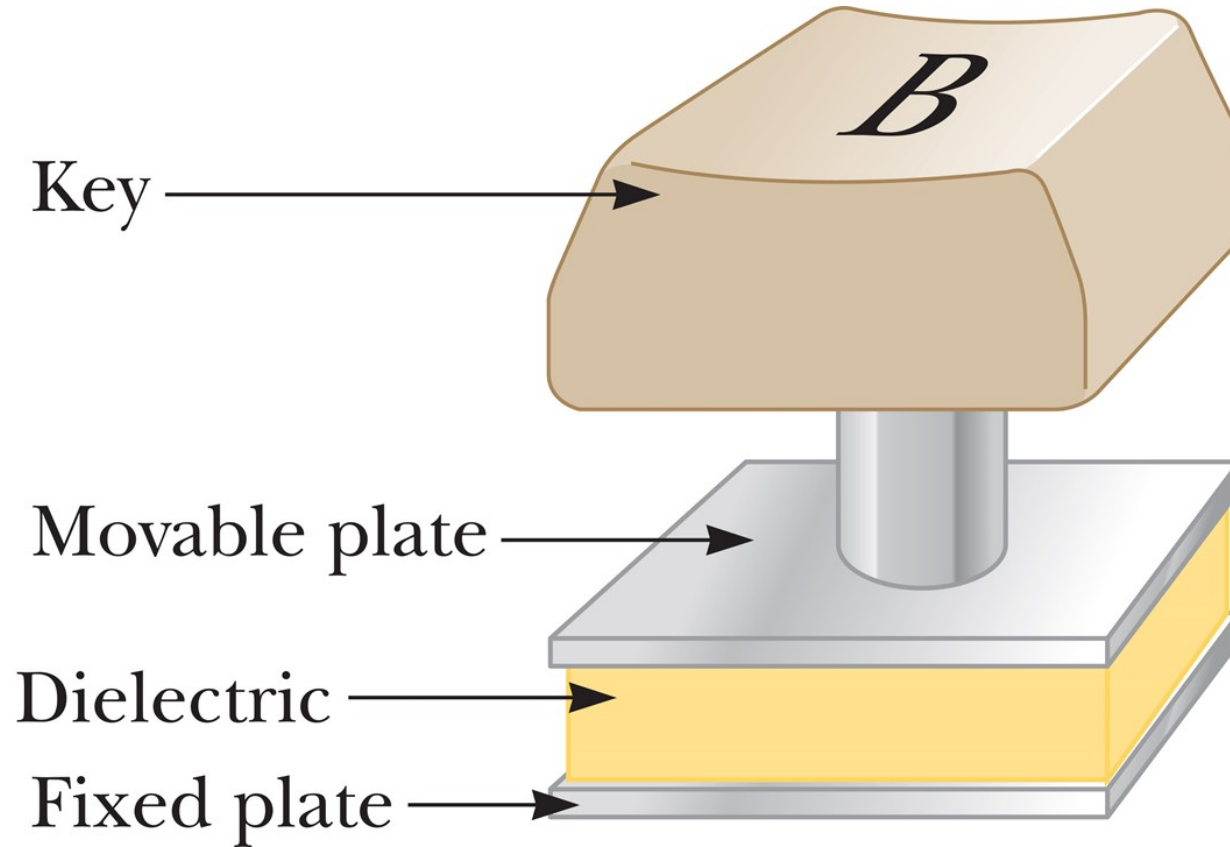
$$C = \frac{Q}{\Delta V} = \frac{\sigma A}{Ed} = \frac{\sigma A}{(\sigma/\epsilon_0)d}$$

$$C = \epsilon_0 \frac{A}{d}$$

The Parallel-Plate Capacitor (2 of 3)



The Parallel-Plate Capacitor (3 of 3)



Symbols for Circuit Elements and Circuits (1 of 2)

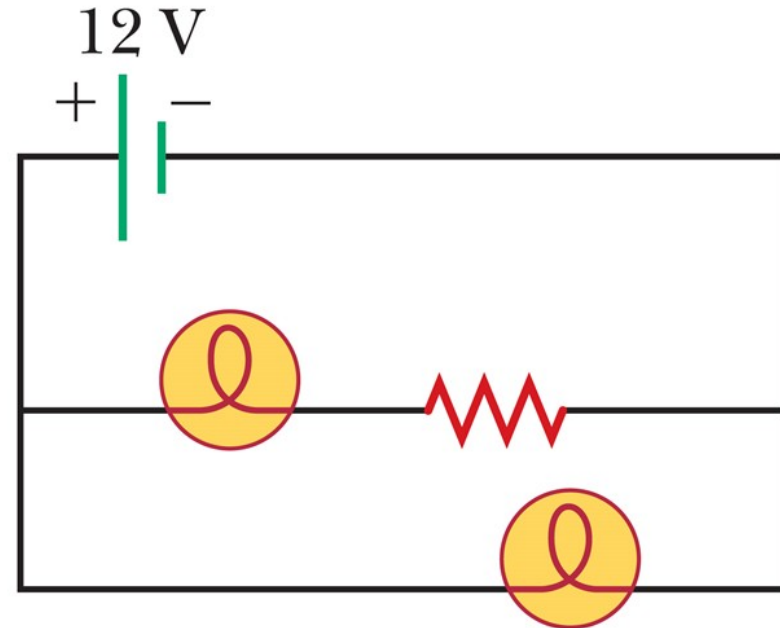
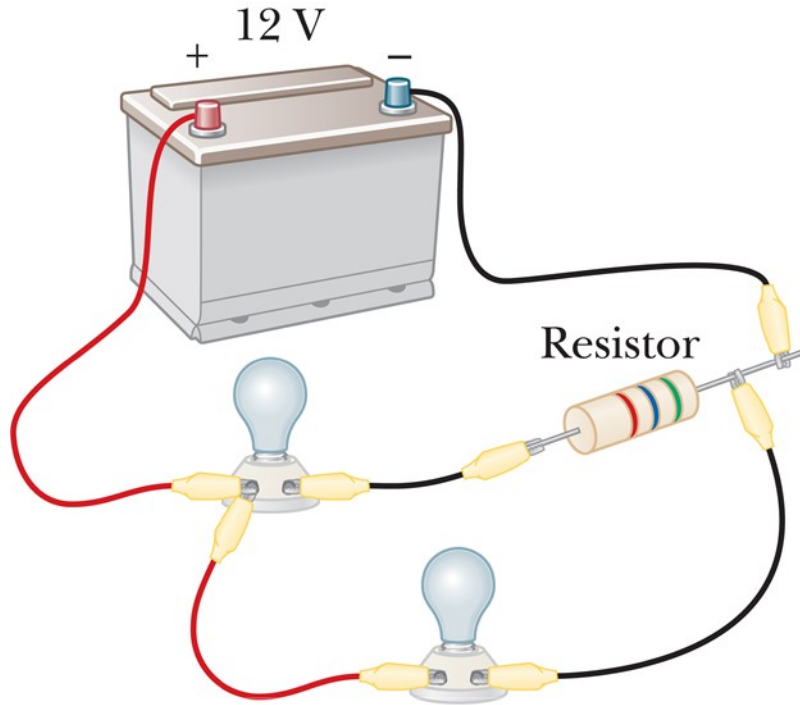
- Capacitor: 

- Battery (or any direct current source): 

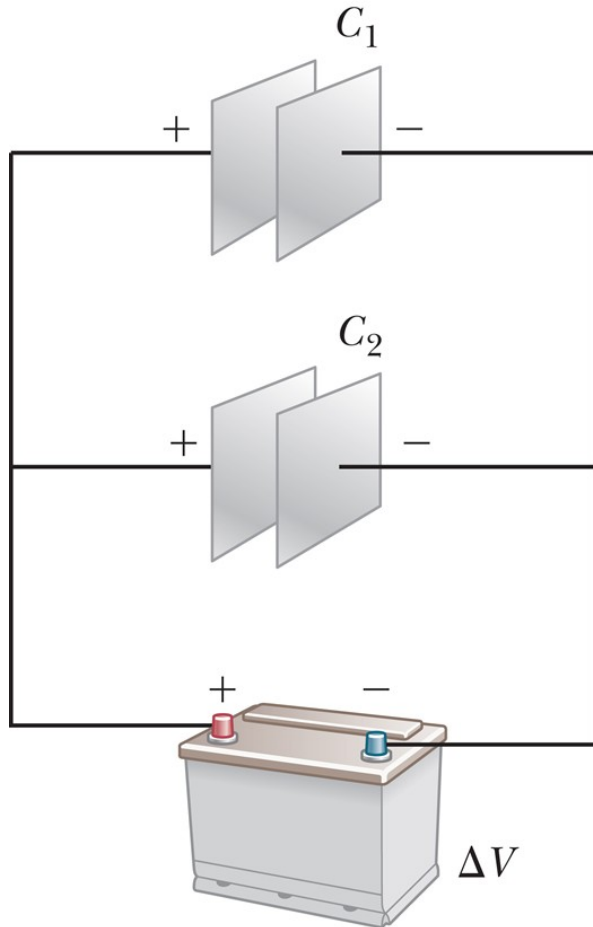
- Resistor: 

- Light bulb: 

Symbols for Circuit Elements and Circuits (2 of 2)



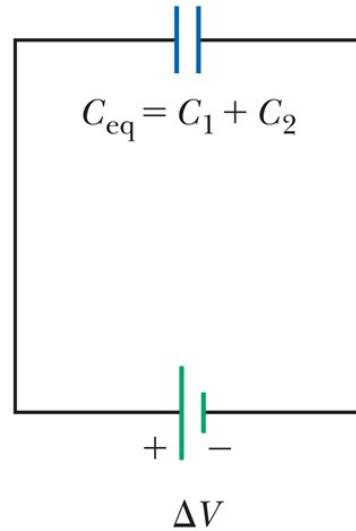
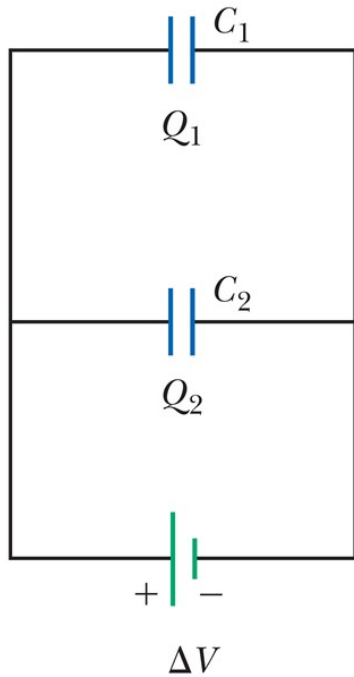
Capacitors in Parallel (1 of 2)



Capacitors in parallel have the same potential difference across them.

Capacitors in Parallel (2 of 2)

$$\Delta V_1 = \Delta V_2 = \Delta V$$



$$Q = Q_1 + Q_2$$

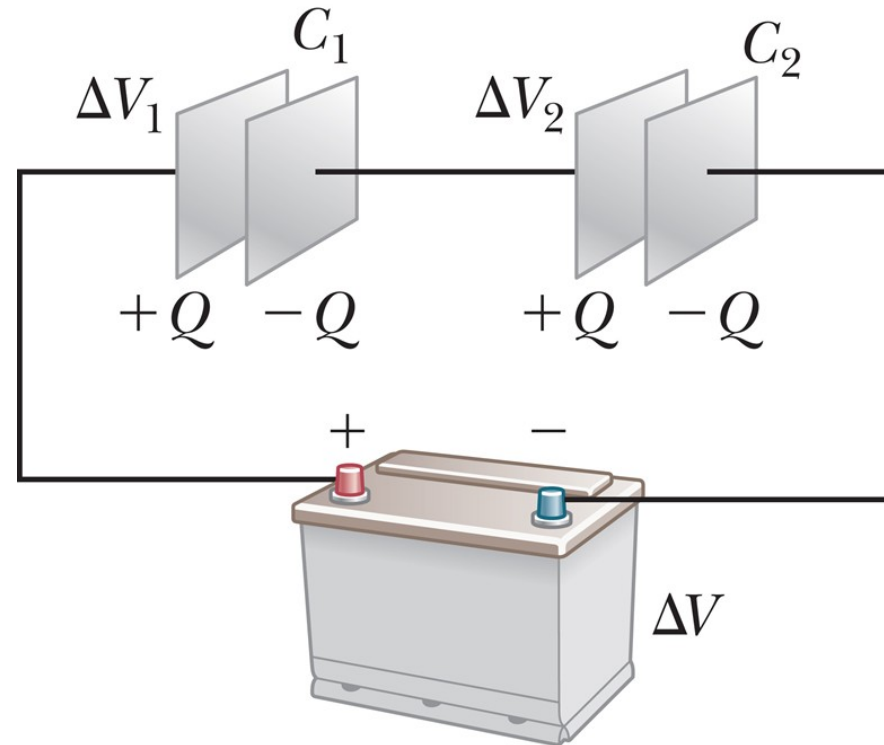
$$Q_1 = C_1 \Delta V \quad \text{and} \quad Q_2 = C_2 \Delta V$$

$$Q = C_{eq} \Delta V$$

$$C_{eq} \Delta V = C_1 \Delta V_1 + C_2 \Delta V_2 \quad \rightarrow \quad C_{eq} = C_1 + C_2$$

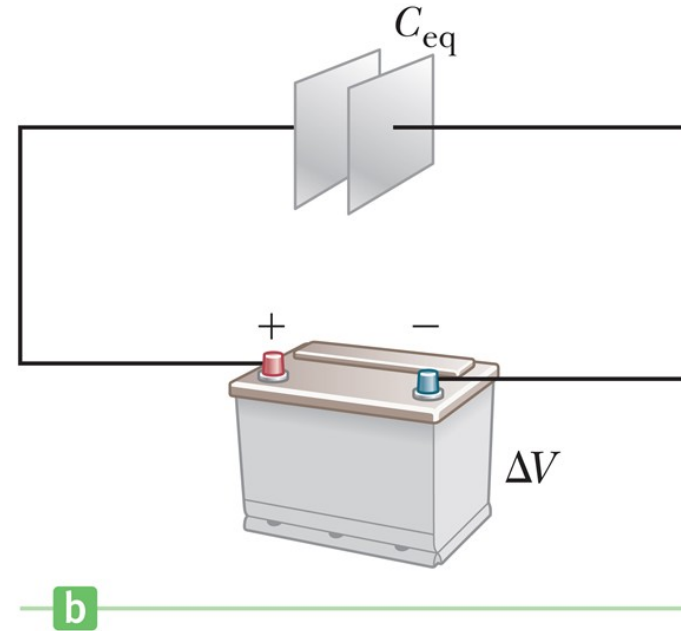
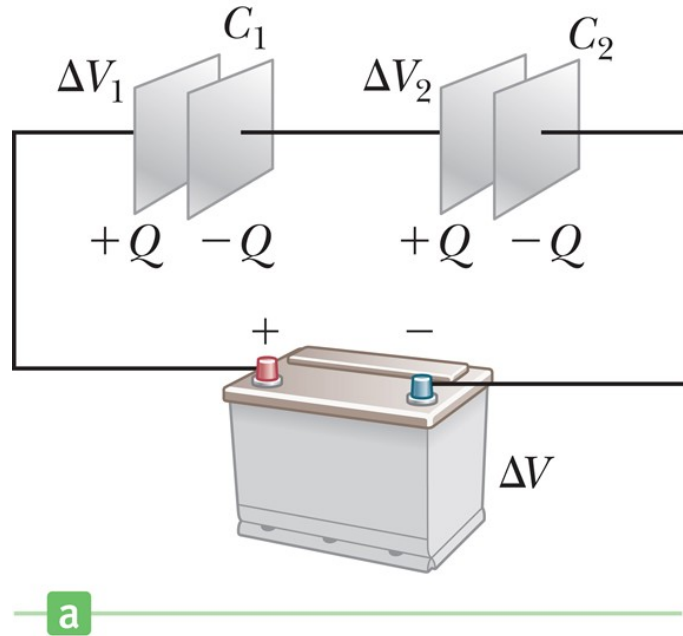
$$C_{eq} = C_1 + C_2 + C_3 + \dots \quad (\text{parallel combination})$$

Capacitors in Series (1 of 3)



For a series combination of capacitors, the magnitude of the charge must be the same on all the plates.

Capacitors in Series (2 of 3)



$$\Delta V = \frac{Q}{C_{\text{eq}}}$$

$$\Delta V_1 = \frac{Q}{C_1}$$

and

$$\Delta V_2 = \frac{Q}{C_2}$$

$$\Delta V = \Delta V_1 + \Delta V_2$$

Capacitors in Series (3 of 3)

$$\Delta V = \Delta V_1 + \Delta V_2$$

$$\frac{Q}{C_{\text{eq}}} = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \quad (\text{series combination})$$

Think – Pair – Share 8

A capacitor is designed so that one plate is large and the other is small. If the plates are connected to a battery,

1. the large plate has a greater charge than the small plate.
2. the large plate has less charge than the small plate.
3. the plates have equal, but opposite, charge.

Problem-Solving Strategy: Complex Capacitor Combinations

1. Combine capacitors
2. Redraw your circuit.
3. Repeat the first two steps
4. Find the charge on the equivalent capacitor: $Q = C \Delta V$
5. Work backward through your diagrams. Use:

A. In parallel:

$$C_{\text{eq}} = C_1 + C_2 + C_3 + \dots \quad \text{with} \quad \Delta V_{\text{eq}} = \Delta V_1 = \Delta V_2 = \Delta V_3 = \dots$$

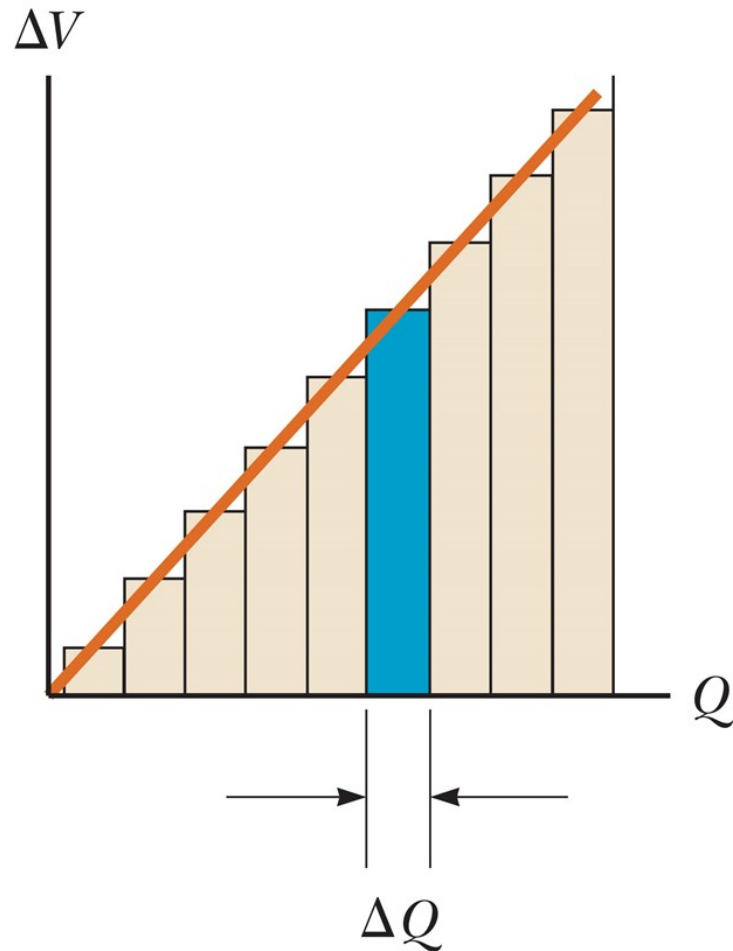
B. In series:

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots \quad \text{with} \quad Q_{\text{eq}} = Q_1 = Q_2 = Q_3 = \dots$$

Energy in a Capacitor (1 of 4)



Energy in a Capacitor (2 of 4)



$$W = -q(V_B - V_A)$$

$$\Delta W = \Delta V \Delta Q$$

$$\Delta V = Q/C$$

$$W = \frac{1}{2} Q \Delta V$$

Energy in a Capacitor (3 of 4)

$$W = \frac{1}{2}Q \Delta V \qquad Q = C \Delta V$$

$$PE_C = \frac{1}{2}Q \Delta V = \frac{1}{2}C(\Delta V)^2 = \frac{Q^2}{2C}$$

$$C = 5.0 \mu\text{F} \quad \Delta V = 120 \text{ V}$$

$$PE_C = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}(5.0 \times 10^{-6} \text{ F})(120 \text{ V})^2 = 3.6 \times 10^{-2} \text{ J}$$

Energy in a Capacitor (4 of 4)



Think – Pair – Share 9

A parallel-plate capacitor is disconnected from a battery, and the plates are pulled a small distance farther apart. Does Q increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

Think – Pair – Share 10

A parallel-plate capacitor is disconnected from a battery, and the plates are pulled a small distance farther apart. Does C increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

Think – Pair – Share 11

A parallel-plate capacitor is disconnected from a battery, and the plates are pulled a small distance farther apart. Does E between the plates increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

Think – Pair – Share 12

A parallel-plate capacitor is disconnected from a battery, and the plates are pulled a small distance farther apart. Does ΔV increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

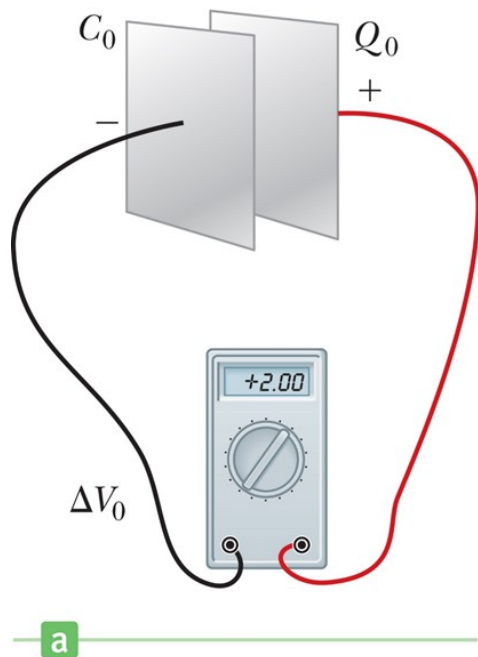
Think – Pair – Share 13

A parallel-plate capacitor is disconnected from a battery, and the plates are pulled a small distance farther apart. Does the energy stored in the capacitor increase, decrease, or stay the same?

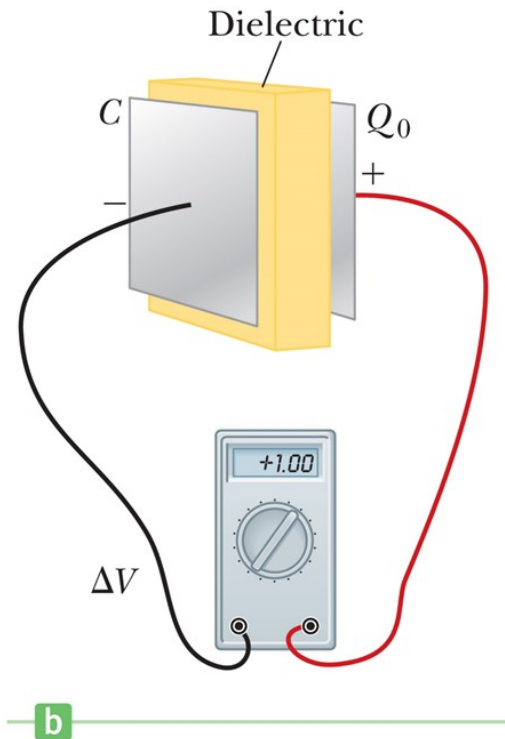
1. increase
2. decrease
3. remains the same

Capacitors with Dielectrics (1 of 5)

Potential difference: ΔV_0
Capacitance: C_0



Potential difference: $\Delta V_0/\kappa$
Capacitance: κC_0



$$\Delta V = \frac{\Delta V_0}{\kappa}$$

Capacitors with Dielectrics (2 of 5)

$$C_0 = \epsilon_0 \frac{A}{d} \quad \rightarrow \quad C = \kappa \epsilon_0 \frac{A}{d}$$

Table 16.1 Dielectric Constants and Dielectric Strengths of Various Materials at Room Temperature

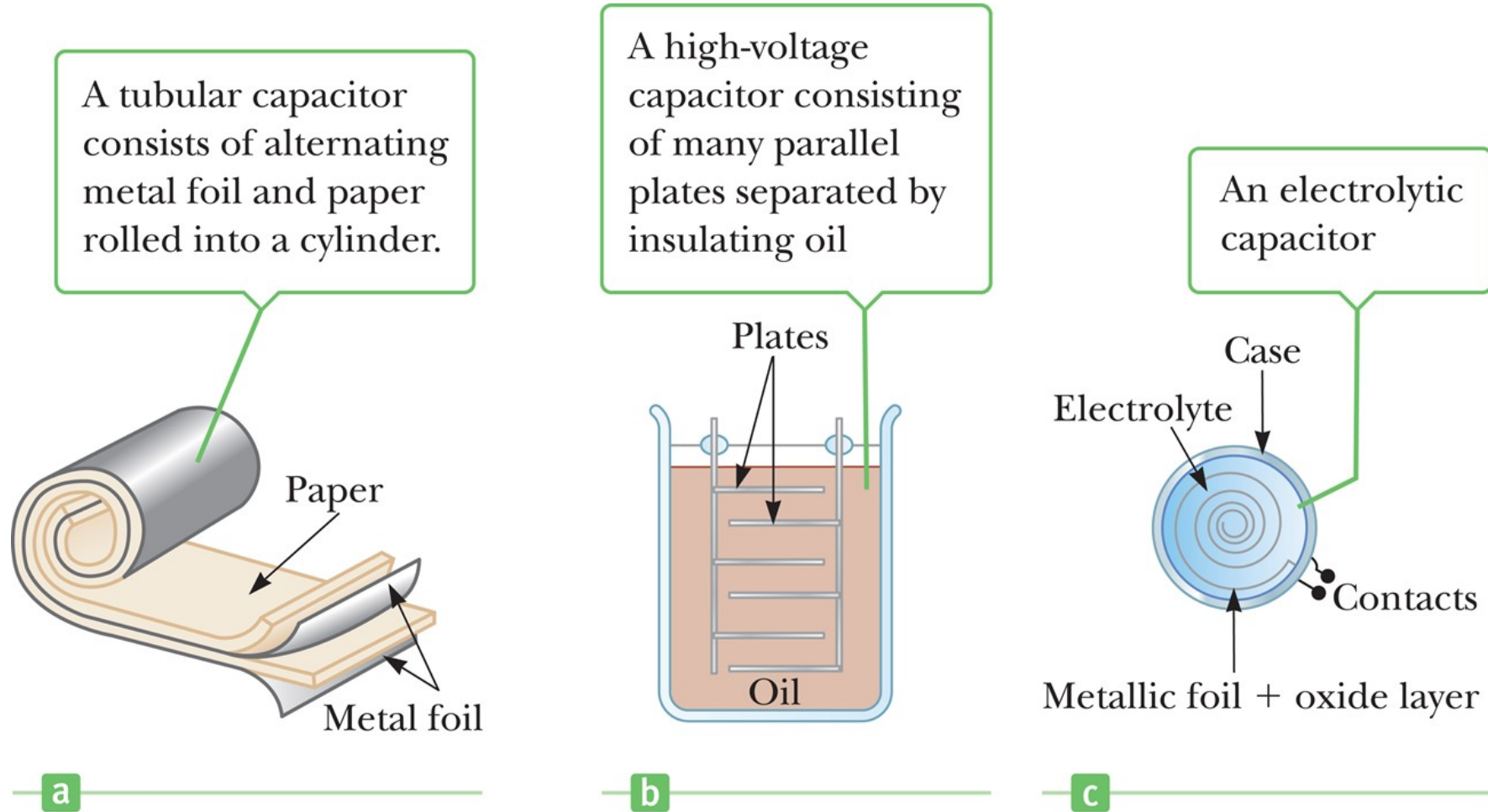
Material	Dielectric Constant κ	Dielectric Strength (V/m)
Air	1.000 59	3×10^6
Bakelite [®]	4.9	24×10^6
Fused quartz	3.78	8×10^6
Neoprene rubber	6.7	12×10^6
Nylon	3.4	14×10^6
Paper	3.7	16×10^6
Polystyrene	2.56	24×10^6
Pyrex [®] glass	5.6	14×10^6
Silicone oil	2.5	15×10^6
Strontium titanate	233	8×10^6
Teflon [®]	2.1	60×10^6
Vacuum	1.000 00	—
Water	80	—

Capacitors with Dielectrics (3 of 5)



Loren Winters/Visuals Unlimited

Capacitors with Dielectrics (4 of 5)

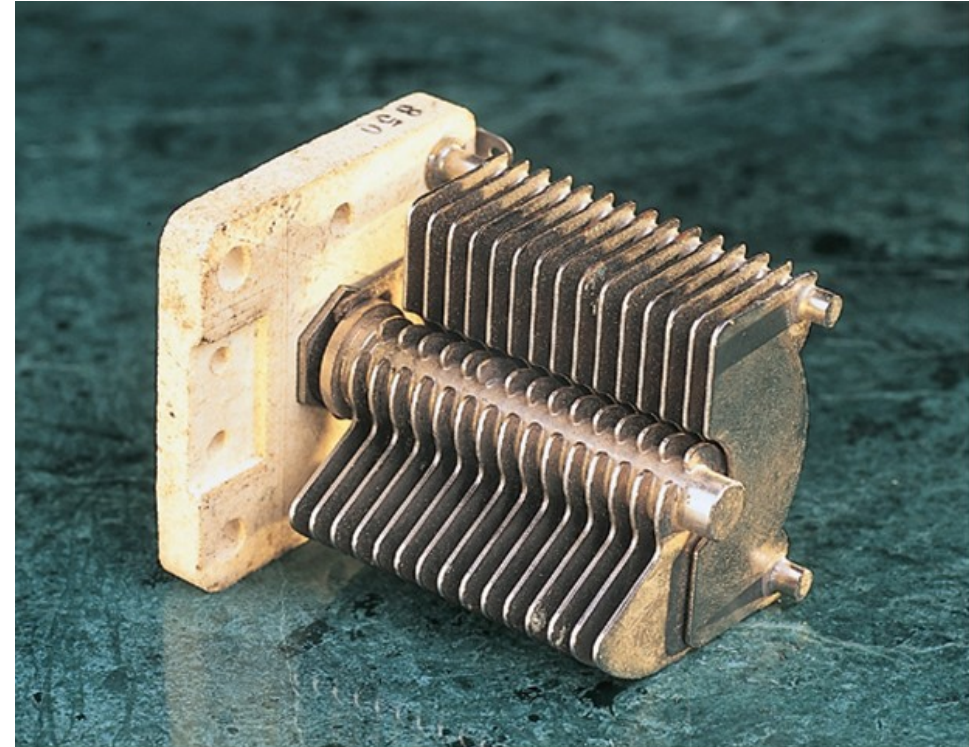


Capacitors with Dielectrics (5 of 5)



Chris Vuille

a



b

© Cengage Learning/George Sample

Think – Pair – Share 14

A fully charged parallel-plate capacitor remains connected to a battery while a dielectric is slid between the plates. Does C increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

Think – Pair – Share 15

A fully charged parallel-plate capacitor remains connected to a battery while a dielectric is slid between the plates. Does Q increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

Think – Pair – Share 16

A fully charged parallel-plate capacitor remains connected to a battery while a dielectric is slid between the plates. Does E between the plates increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

Think – Pair – Share 17

A fully charged parallel-plate capacitor remains connected to a battery while a dielectric is slid between the plates. Does ΔV increase, decrease, or stay the same?

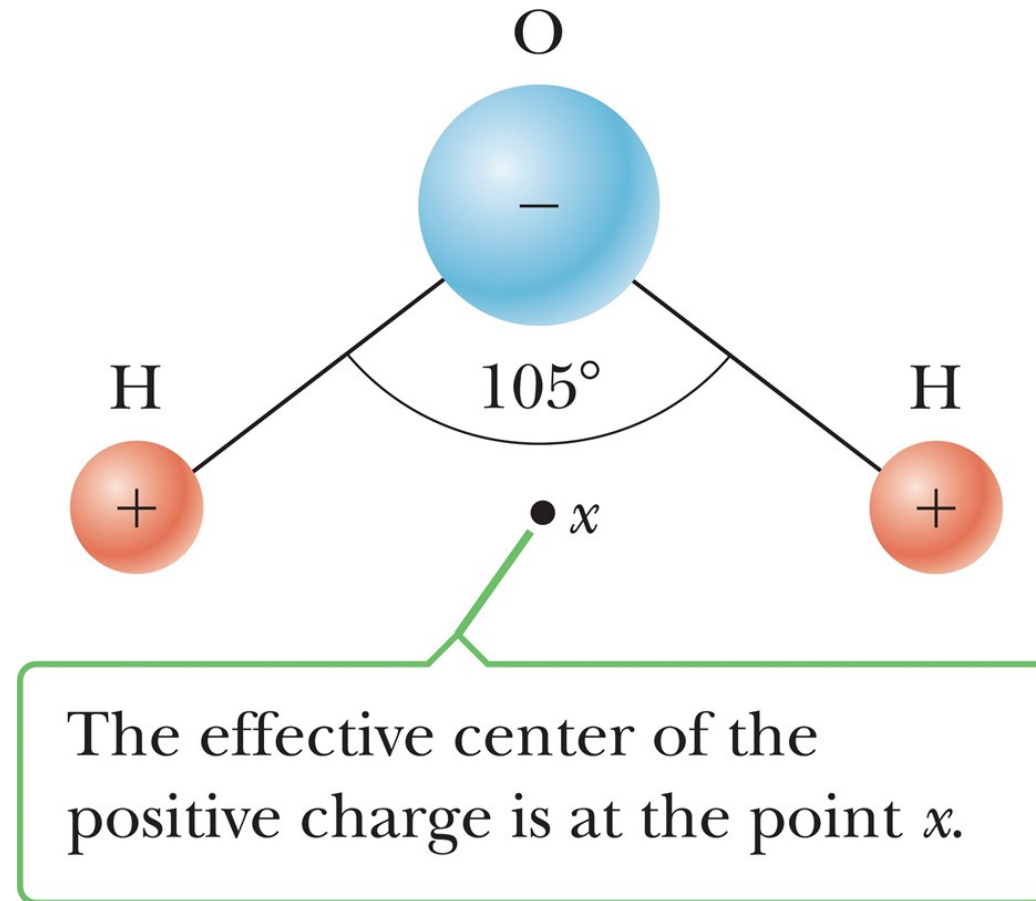
1. increase
2. decrease
3. remains the same

Think – Pair – Share 18

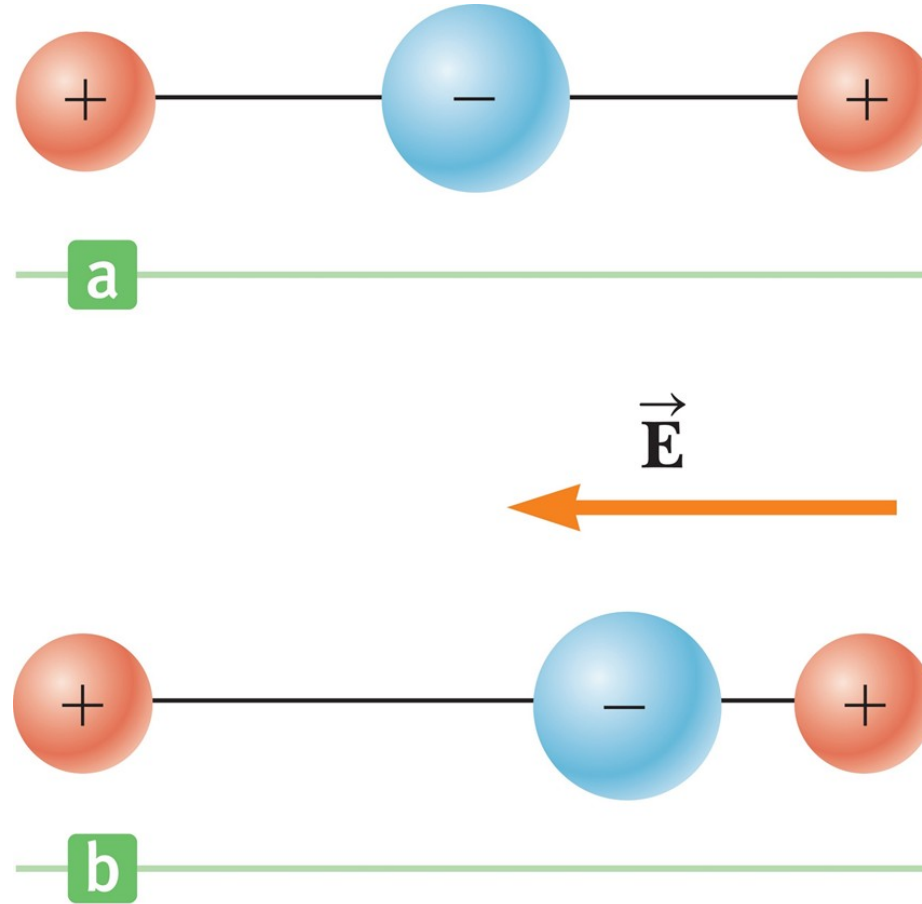
A fully charged parallel-plate capacitor remains connected to a battery while a dielectric is slid between the plates. Does the energy stored in the capacitor increase, decrease, or stay the same?

1. increase
2. decrease
3. remains the same

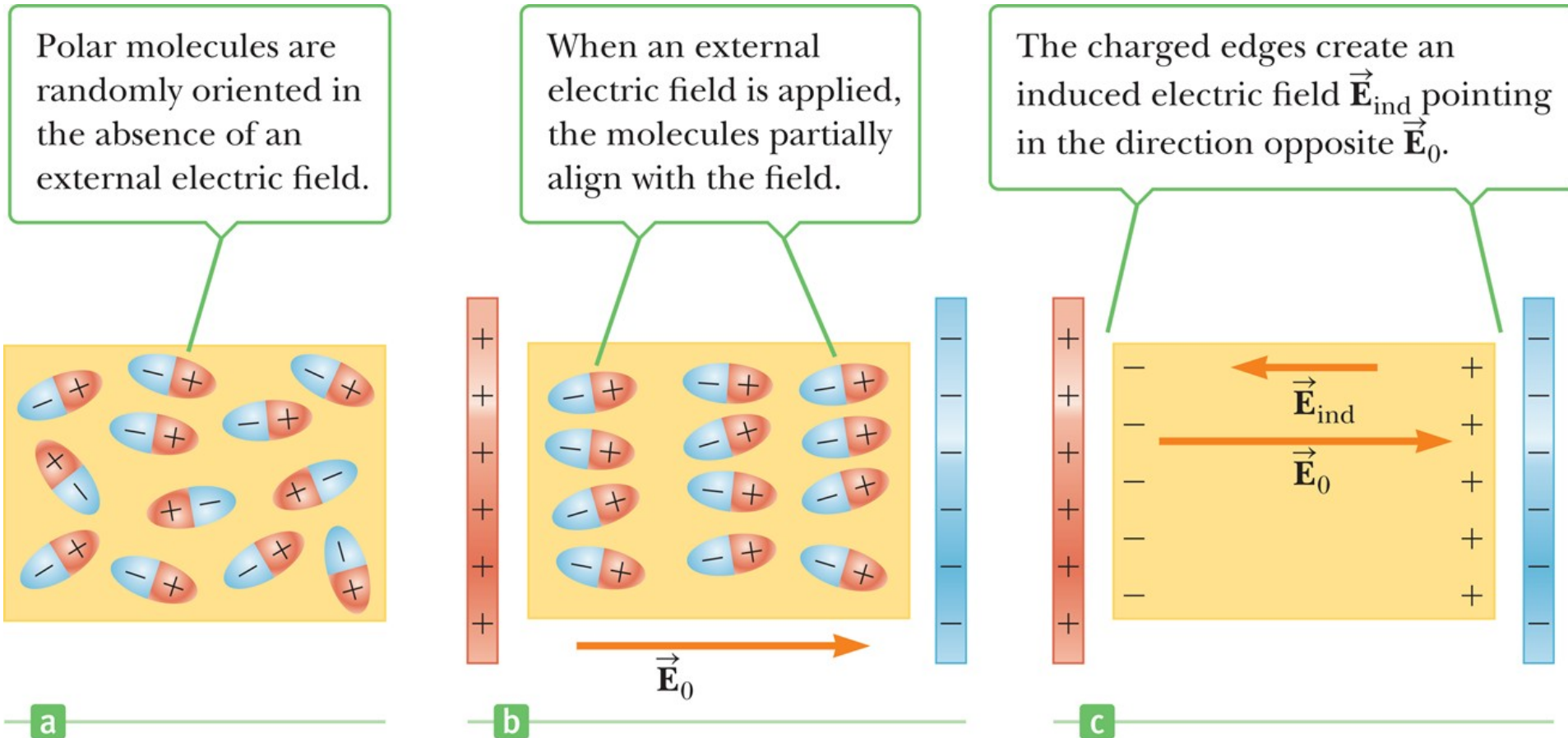
An Atomic Description of Dielectrics (1 of 3)



An Atomic Description of Dielectrics (2 of 3)



An Atomic Description of Dielectrics (3 of 3)



Think – Pair – Share 19

Consider a parallel-plate capacitor with a dielectric material between the plates. If the temperature of the dielectric increases, the capacitance

1. increase
2. decrease
3. remains the same

Think – Pair – Share 20

The electric potential at two points in space is: $V_1 = 200$ volts and $V_2 = 300$ volts. Which of the following statements is true for moving a point charge q from point 1 to 2?

- A. The work done by an external agent to move q from point 1 to 2 is positive.
- B. We can't determine the work done because we don't know the direction of V at the two points.
- C. The work done by the electric force exerted on q in moving it from point 1 to 2 is $W = -q(100 \text{ V})$.

- 1. A
- 2. B
- 3. C
- 4. A and B
- 5. A and C
- 6. B and C
- 7. A, B, and C
- 8. None of the above